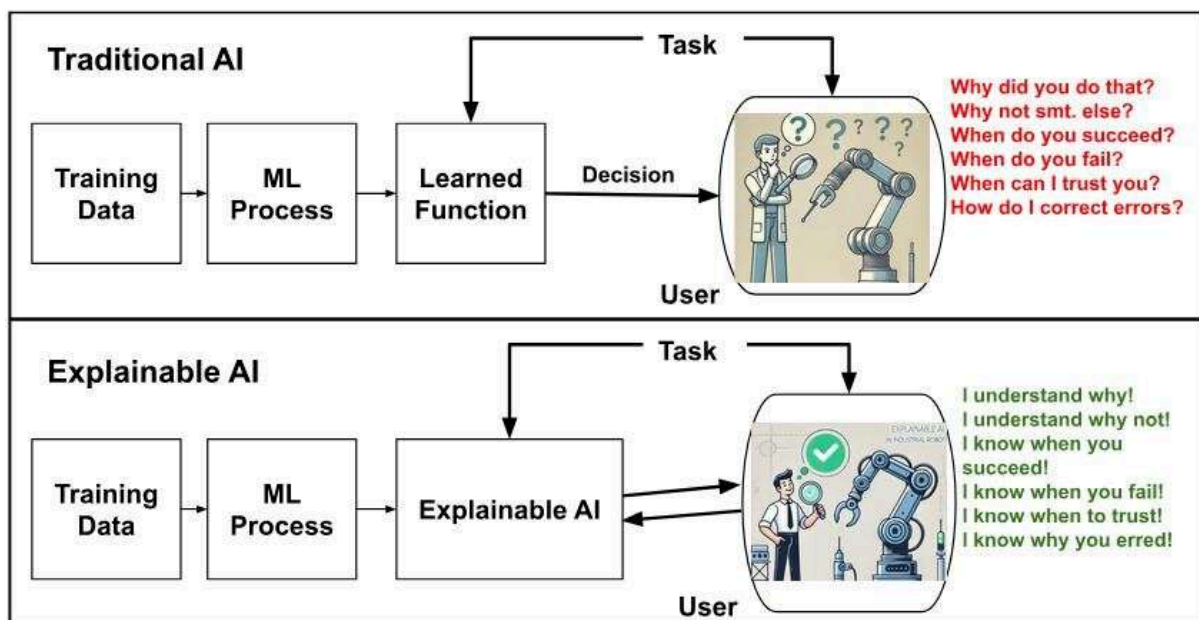


What is Explainable AI

Machine learning is now used in many parts of our daily life, products, and research. But most AI systems cannot clearly explain how they make decisions or predictions. Because of this, people may not fully trust the results, and sometimes errors can stay hidden. Explainable AI (XAI) helps solve this problem. It helps people understand how an AI system works and why it gives a certain result. XAI makes complex AI models easier to understand, trust, and improve.

In simple terms:

Traditional AI tells us the answer. XAI tells us the answer and the reason behind it.



Why Do We Need XAI?

AI and machine learning are now used in many important areas, but most AI systems do not explain how they make decisions. Because of this, people may find it hard to trust the results.

Explainable AI (XAI) helps people **understand how AI systems work** and **why they give certain results**. It also helps users find mistakes and trust the system more.

Today, clear explanations are very important in fields like healthcare, banking, and the military, where decisions can seriously affect people's lives.

The Importance of Explainable ai

One of the main goals of Explainable AI (XAI) is to **make AI systems more accountable and trustworthy**. In the past, many AI systems worked like “black boxes.” This means people could see the input and the final result, but they could not understand how the AI actually made its decisions. The algorithms were often too complex, hidden, or difficult for normal users to understand. Explainable AI helps solve this problem by making AI decisions clearer and easier for humans to understand. AI is now used in areas like healthcare and banking, so it is important that these systems make fair and trusted decisions. AI should not treat people unfairly because of bias. For example, someone should not be refused a loan for unfair reasons.

In healthcare, an AI system trained with limited data may not correctly find diseases in people from different races, genders, or countries. That is why people need to understand how the AI system makes decisions. This helps find mistakes and reduce bias.

As AI is becoming more common, it is very important to make sure AI systems are fair, honest, and trustworthy.

BUILDING TRUST

Trust is essential for AI adoption

Doctors, bankers, and engineers are more likely to rely on AI systems when explanations are available.

Example

Cancer Prediction:

Without XAI: **Cancer Detected**

With XAI:

Cancer Detected because:

- Tumor Size = 45%

- Cell Density = 35%

- Irregular Shape = 20%

Detecting Bias

AI systems can learn harmful biases from historical data.

Example:

A hiring system trained on biased data may discriminate against certain groups.

XAI helps reveal such biases.

Regulatory compliance

Many governments require transparent AI systems.

Important principles include:

- Accountability
- Fairness
- Transparency
- Reliability

Improving Model Performance

Developers can identify weaknesses in AI models through explanations.

Interpretability and Explainability

Interpretability and Explainability are both important in AI, but they are slightly different from each other. The following table shows their main differences clearly.

Aspect	Interpretability	Explainability
Focus	Focused on understanding how the model works inside	Focuses on explaining individual results after the decision is made.
Scope	Can be applied to the whole model or to a specific part of it.	Usually explains a single prediction or output.
Limitations	Interpretable models may become simpler and easier to understand, but sometimes their performance can be lower	Explanations can sometimes be confusing if they make a complex model look too simple.
Complexity	works best for simpler models	More useful for complex “black-box” models like deep learning.
Benefits	Interpretable models are easier to find mistakes in and improve because their process is clear and easy to understand.	Helps people understand AI decisions without knowing all the technical details. It also supports privacy and AI rules like GDPR.
Example	A teacher using a simple model to calculate grades.	A teacher using a complex AI tool to create personalized learning material.

Global Explainability

Global explainability describes the overall behavior of a machine learning model.

Example

A loan approval model may generally prioritize:

- Credit Score = 60%
- Income = 25%
- Employment Status = 15%

This explanation describes the model's overall decision-making strategy.

Local Explainability

Local explainability focuses on a specific prediction.

Example:

For a particular customer:

Loan Approved because:

- Credit Score = +50%
- Income = +30%
- Existing Debt = -10%

This explains why the model made that specific decision.

The Black Box Problem

At the center of Explainable AI (XAI) is something called the *black box problem*. The most powerful AI models, like deep neural networks and gradient boosting models, are also the hardest to understand. Even the people who build them often cannot clearly explain how a specific decision was made. This creates a serious issue. A model may look very accurate on test data, but it might be making predictions for the wrong reasons. For example, it may correctly classify a document about Christianity or Atheism, but actually rely on unrelated words like “Host” or “NNTP” that just happened to match the training data. Without XAI, this kind of hidden mistake can go unnoticed. Studies show there is a trade-off between accuracy and interpretability. Simple models like decision trees are easy to understand, but they may not be very accurate. On the other hand, complex models are highly accurate but difficult to explain. The main goal of XAI is to reduce this gap by either making models more understandable or by explaining complex models after they make predictions.

The Black Box Problem

AI models can generally be divided into two categories:

White -Box Models

White-box models are naturally interpretable.

Examples:

- Linear Regression
- Logistic Regression
- Decision Trees
- Rule-Based Systems

Users can directly understand how predictions are generated.

Example:

Decision Tree: The reasoning is clearly visible.

Black-Box-Models

Black-box models are difficult to interpret because they contain millions or even billions of parameters.

Examples:

- Deep Neural Networks
- Convolutional Neural Networks (CNNs)
- Vision Transformers (ViTs)
- Large Language Models (LLMs)

Although these models often provide superior performance, understanding their internal decision process is extremely difficult.

Example

Input:

Customer Information

Output:

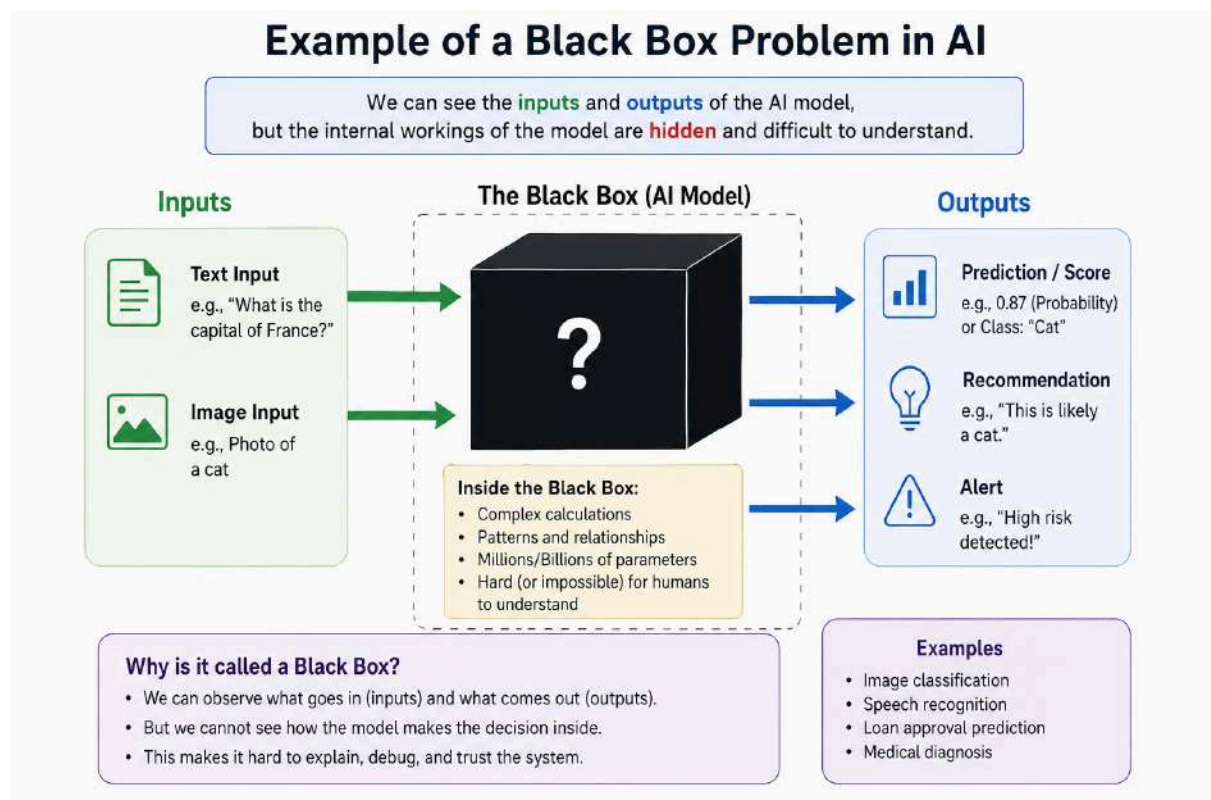
Loan Rejected

No explanation is provided regarding why the loan was rejected.

The Black Box Problem in AI

We can see the inputs and outputs, but **the internal workings of the model are hidden and difficult to understand.**

The figure clearly shows the black box problem in AI. On the left side, different inputs like text, images, and data are given to the AI model. In the middle, the AI works like a “black box,” where we cannot see or clearly understand what is happening inside. On the right side, we only get the final results, such as predictions, decisions, or recommendations. This means we can see what goes in and what comes out, but we do not know the exact process inside the model. Because of this, it is difficult to explain why the AI made a specific decision or how it produced that result.



How XAI Methods Are Categorized

XAI (Explainable AI) methods are usually divided into **three main groups** to make them easier to understand and use.

Scope :

Some methods explain the whole model , while others explain one specific decision.

LIME and SHAP - local methods (explain one result)

Permutation Feature Importance -

global method (explains overall model behavior)

Timing -

This shows *when* the explanation is added.

Ante-hoc methods -

built into the model from the beginning (easy to understand by design)

Example: Decision Trees

Post-hoc methods -

applied after training the model most XAI methods fall here because advanced models are usually complex and act like a “black box”

Model Dependency -

This tells whether the method depends on a specific model or not.

Model-agnostic methods -

work on any AI model Flexible and widely used

Model-specific methods -

designed for a particular type of model more accurate for that model but not widely reusable.

Notable Xai Methods

Lime

LIME helps explain one prediction at a time in a simple way. It slightly changes the input data and checks how the AI model reacts. Then, it creates an easy explanation for that specific prediction.

One big advantage of LIME is that it can work with almost any AI model.

Shape

SHAP shows how much each feature affected the final prediction. It explains which features increased or decreased the result.

Today, LIME and SHAP are two of the most popular XAI methods.

Anchors

Anchors explain AI decisions using simple If- Then Rules. These rules show when the model will keep making the same prediction with high confidence.

Because the rules are easy to read, people can understand them quickly.

Example:

If attendance is high and marks are good, then the student will pass.

Layer-wise Propagation (LP)

LP is mainly used for neural networks. It traces the prediction backward through the network and finds which input features had the biggest impact on the final result.

Grad-CAM

Grad-CAM is mostly used for image-based AI systems. It highlights the parts of an image that influenced the model's decision the most. Usually, the explanation appears **as a heatmap** on the image.

Example:

In a medical scan, Grad-CAM can show the area that helped the AI detect a disease.

Mathematical Foundation of Explainable AI

One of the most popular mathematical approaches in XAI is the Shapley Value from Cooperative Game Theory.

Shapley Value Formula:

$$\phi_i = \sum_{s \subseteq N/\{i\}} \frac{|s|!(n-|s|-1)!}{n!} [f(s \cup \{i\}) - f(s)]$$

Where:

- ϕ_i = Contribution of feature i
- S = Subset of features
- n = Total number of features
- $f(S)$ = Model prediction using subset S

The formula calculates the average contribution of each feature across all possible feature combinations.

Example of Shapley Value

$$\phi_i = \sum_{s \subseteq N/\{i\}} \frac{|s|!(n-|s|-1)!}{n!} [f(s \cup \{i\}) - f(s)]$$

Suppose a loan approval model uses:

- Income
- Credit Score

Prediction Score = 100

Contributions:

<i>Feature</i>	<i>contribution</i>
Location	\$100,000

Area	\$80,000
Bedrooms	\$40,000
Age	\$30,000

Total Prediction: 40+60=100

Thus, Credit Score contributed more to the decision.

Popular Explainable AI Techniques

1. LIME (Local Interpretable Model-Agnostic Explanations)

LIME explains individual predictions by creating a simpler interpretable model around a specific prediction.

Formula

$$\text{Explanation}(x) = \arg \min_{g \in G} L(f, g, \pi_x) + \Omega(g)$$

Where:

- f = Original model
- g = Simple interpretable model
- L = Loss function
- π_x = Local neighborhood
- $\Omega(g)$ = Complexity penalty

Working Process

1. Select an instance.
2. Generate neighboring samples.
3. Obtain predictions.
4. Train a simple interpretable model.
5. Explain the prediction.

Example

Sentence:

"The movie was fantastic and inspiring."

Prediction:

Positive = 97%

<i>word</i>	<i>contribution</i>
Fantastic	+0.55
Inspiring	+0.32
Movie	+0.04

2. SHAP (SHapley Additive Explanations)

SHAP uses **Shapley values** to determine feature importance.

Formula:
$$f(x) = \phi_0 + \sum_{i=1}^M \phi_i$$

Where:

- ϕ_0 = Base value
- ϕ_i = Feature contribution

Example

House Price Prediction

Base Price = \$100,000

Contributions:

- Area = +\$30,000
- Location = +\$20,000

Final Price:

100000+30000+20000

=150000=150000=15000

3.Attention Visualization

Attention mechanisms help identify which parts of the input influence the model's decision.

Example

Image Classification

The model identifies a cat by focusing on:

- Face
- Eyes
- Ears

Heatmaps show which regions contributed most.

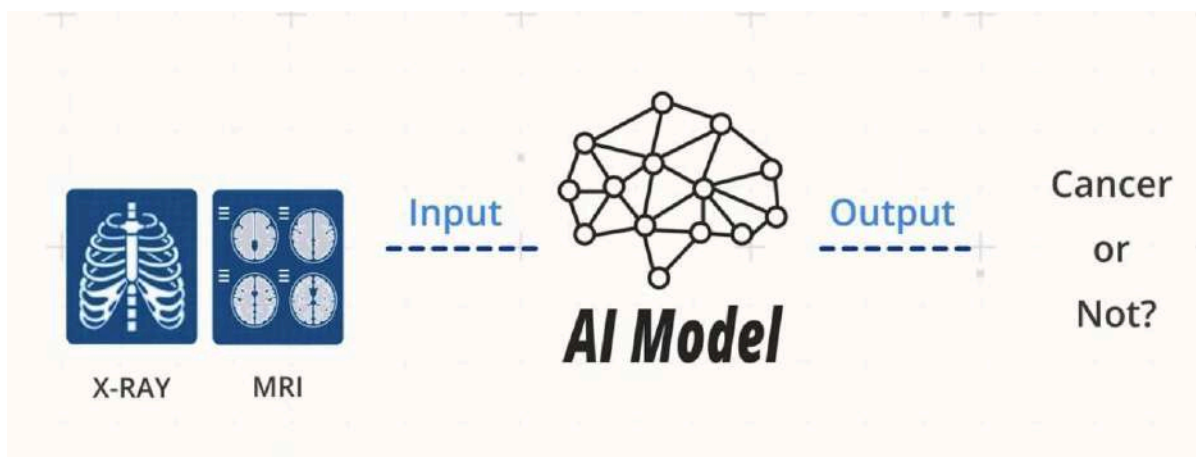
Where XAI is Applied

Explainable AI (XAI) is used in many important fields where understanding AI decisions is necessary.

Healthcare

In healthcare, AI helps doctors detect diseases such as tumors from medical images and scan reports. But doctors cannot fully trust a result without knowing the reason behind it.

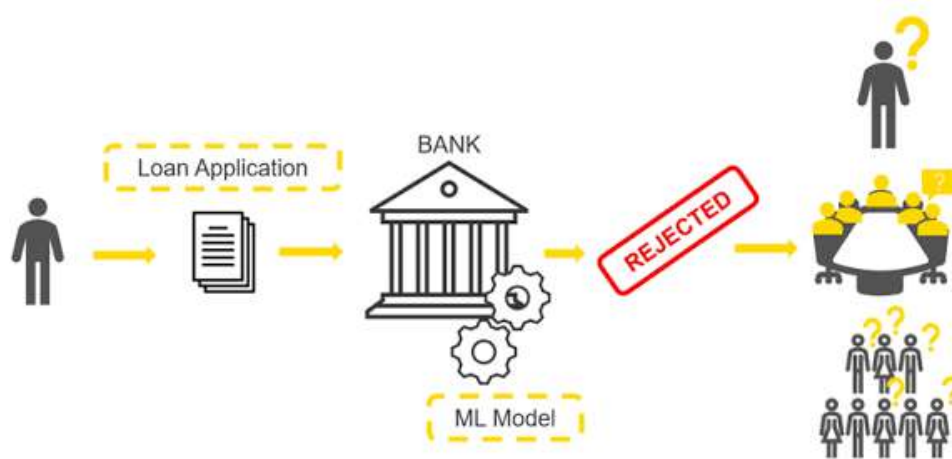
XAI helps by showing which parts of the scan influenced the prediction. This makes medical decisions safer and more reliable.



Banking and financial

Banks and financial companies use AI for credit scoring and loan approval. These decisions can strongly affect people's lives and financial opportunities.

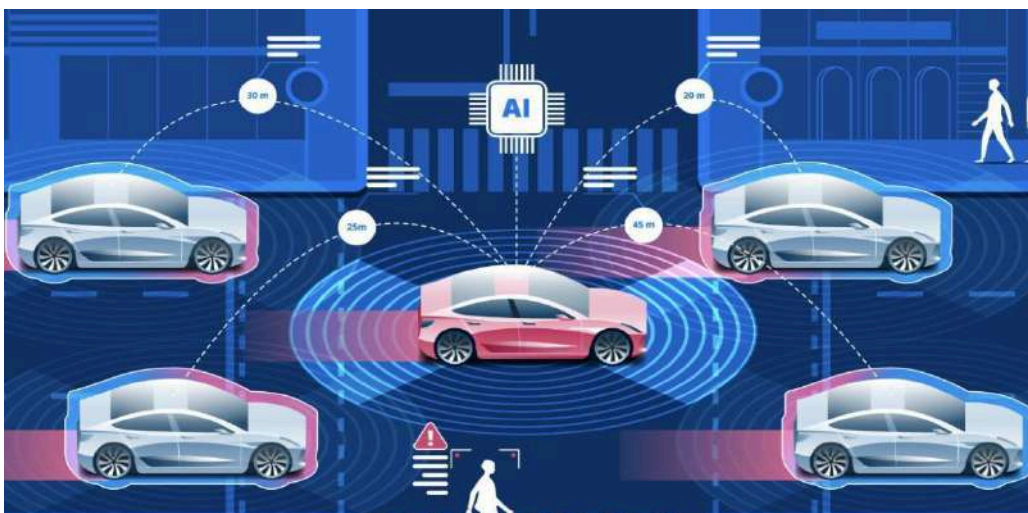
In many countries, institutions must explain why a loan application was rejected. XAI helps provide clear and understandable explanations for these decisions.



Autonomous Driving

Self-driving cars also use AI to make decisions like braking, turning, or avoiding obstacles. For safety reasons, engineers need to understand why the system made a certain decision.

XAI helps analyze the model's behavior in different situations and ensures that the vehicle is acting correctly and safely.



Benefit of Explainable AI

Improved Trust

Users can understand model decisions.

Better Decision-Making

Human experts can verify AI recommendations.

Fairness

Biases can be detected and corrected.

Regulatory Compliance

Supports legal and ethical requirements.

Easier Debugging

Developers can identify model errors more effectively.

Limitations of Current XAI Methods

Although Explainable AI (XAI) has many benefits, it still has some important limitations and challenges.

1. Oversimplification

Sometimes XAI methods make complex AI models look too simple. Because of this, the explanation may not fully show how the model actually works.

A simple explanation can be easier to understand, but it may miss some important details of the real decision-making process.

2. Lower Model Performance

In many cases, XAI systems may not perform as well as black-box AI models.

Highly accurate models are often more complex and difficult to explain, while simpler and explainable models may give lower prediction accuracy.

3. Training Complexity

Building an AI system that can also explain its decisions is more difficult than building a normal black-box model. It usually needs extra training, design, and computational resources, which makes the development process more complicated.

4. Privacy Risks

Some XAI systems work with sensitive or confidential data. Since XAI focuses on transparency, there is a chance that private information could be revealed through the explanations. This creates concerns about data privacy and security.

5. Different People Need Different Explanations

Not everyone understands explanations in the same way. A developer, doctor, regulator, or normal user may all need different types of explanations from the same AI system. Because of this, creating one explanation that satisfies everyone is difficult.

6. Some Methods Are Not Fully Reliable

Some XAI methods can produce different explanations for the same prediction. For example, small changes in LIME may create slightly different results, which can reduce consistency and trust.

7. Explanations Are Not Always the Full Truth

Most XAI methods only provide approximate explanations instead of showing the exact internal process of the AI model. As a result, users may sometimes misunderstand how the model actually works.

Conclusion:

Explainable Artificial Intelligence (XAI) is a crucial advancement in the development of trustworthy and responsible AI systems. While modern machine learning models provide exceptional predictive performance, their lack of transparency creates significant challenges in critical applications. XAI bridges the gap between machine intelligence and human understanding by making AI decisions transparent, interpretable, and accountable. Techniques such as SHAP, LIME, Saliency Maps, Grad-CAM, Counterfactual Explanations, and Attention Visualization enable users to understand how AI systems arrive at their conclusions. As AI continues to influence important aspects of society, explainability will become a fundamental requirement rather than an optional feature. Future AI systems are expected to combine high accuracy with meaningful explanations, ensuring fairness, trust, safety, and ethical deployment across diverse domains.

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